

1.3 #s: 3, 5, 7, 10

1.4 #s: 1, 2, 7, 10, 12

1.5 #s: 2, 3, 6

Homework 2 (1.3, 1.4, 1.5), Math 3210, Tyler Treth

3. Prove that if $\mathbb{Z}$ satisfies the axioms for a commutative ring, then $\mathbb{Q}$ satisfies A1 and M1.

Since $\mathbb{Z} \subset \mathbb{Q}$, it suffices to show proofs for only those elements $x \in \mathbb{Q}$ s.t. $x \notin \mathbb{Z}$.

Thus, elements $x \in \mathbb{Q}$ are of the form $\frac{p}{q}$, $p, q \in \mathbb{Z}$, where $q \neq 0$ and $p \neq 0$ and $p \nmid kq$ for any $k \in \mathbb{Z}$.

A1: Two elements of the aforementioned form, x, y , $x + y = y + x$ for all $x, y \in \mathbb{Q}$.

Proof:

Say $x = \frac{p}{q}$ and $y = \frac{r}{l}$, for any $p, q, r, l \in \mathbb{Z}$ with properties mentioned above. Then $\frac{p}{q} + \frac{r}{l}$, defined to be $\frac{pl + qr}{ql}$. Since $p, q, r, l \in \mathbb{Z}$ which satisfy the axioms

of a commutative ring, then $\frac{pl + qr}{ql} = \frac{qr + pl}{ql}$ (A2) $\frac{qr + pl}{ql}$ (M1) $\frac{rq + lp}{lq}$ which, by definition, is $\frac{r}{l} + \frac{p}{q}$. Thus, $\frac{p}{q} + \frac{r}{l} = \frac{r}{l} + \frac{p}{q}$, $x + y = y + x$. $\square$

M1: Two elements of the aforementioned form, x, y , $xy = yx$ for all $x, y \in \mathbb{Q}$.

Proof:

Say $x = \frac{p}{q}$ and $y = \frac{r}{l}$, then $xy = \frac{p}{q} \cdot \frac{r}{l}$ which is, by definition, $\frac{pr}{ql}$. Since $p, q, r, l \in \mathbb{Z}$ and therefore satisfy the axioms of $\mathbb{Z}$ a commutative ring, then $\frac{pr}{ql} = \frac{rp}{lq}$ (M1) which is, by definition, $\frac{r}{l} \cdot \frac{p}{q} = yx$. So, $xy = yx$. $\square$

1.3 #s: 5, 7, 10

1.4 #s: 1, 2, 7, 10, 12

1.5 #s: 2, 3, 6

Homework 2, cont.

5. Prove $(-x)(-y) = xy$.

Lemma: $(-x) = (-1)x$ Lemma 2: $(-1)(-1) \stackrel{M2}{=} 1$
Proof: $(-x)(-y) \stackrel{\text{lemma}}{=} (-1)(x)(-1)(y) \stackrel{M2}{=} (-1)(-1)(x)(y)$
 $\stackrel{M2}{=} 1(x)(y) \stackrel{\text{lemma 2}}{=} xy \quad \square$

Lemma Proof: $(-x) = (-1)x$, add x to both sides $(-x) + x = (-1)x + x$

LHS: $(-x) + x \stackrel{A4}{=} 0$
RHS: $(-1)x + x \stackrel{M3}{=} (-1)x + (1)x \stackrel{M1}{=} x(-1) + x(1) \stackrel{D}{=} x(-1+1)$
 $\stackrel{A4}{=} x(0) \stackrel{\text{ex 1.3.2}}{=} 0$. So, $0=0$. $\square$

Lemma 2 Proof: $(-1)(-1) = 1$, add (-1) to both sides.

$(-1)(-1) + (-1) = 1 + (-1)$
LHS: $(-1)(-1) + (1)(-1) \stackrel{M2}{=} (-1)(-1) + (-1)(1) \stackrel{D}{=} (-1)(-1+1)$
 $\stackrel{A4}{=} (-1)(0) \stackrel{\text{ex 1.3.2}}{=} 0$
RHS: $1 + (-1) \stackrel{A4}{=} 0$. So, $0=0$. $\square$

7: $xy=0$ implies $x=0$ or $y=0$.

Proof: Suppose $xy=0$ is true.

Proceed by cases: Lemma 1.3.2. $x \cdot 0 = 0$.

Case 1: $x=0$.

$xy=0$ and $x=0$. Then $0y=0$; by M2 $y \cdot 0 = 0$ is true by lemma.

Case 2: $y=0$.

$xy=0$ and $y=0$. Then $x0=0$; true by lemma.

Case 3: $x=0, y=0$

$xy=0$ and $x=y=0$. Then $00=0$; true by lemma. $\square$

Lemma 1.3.2 Proof: $x \cdot 0 = 0$. Add x to both sides
 $x \cdot 0 + x = 0 + x$, by A3, $x \cdot 0 + (1)x = 0 + x$, by M2,
 $x \cdot 0 + x(1) = 0 + x$, by D $x(0+1) = 0 + x$, by A3, $x(1) = x$
which is true by M2 and M3. $\square$

1.3 #s: 10

1.4 #s: 1, 2, 7, 10, 12

1.5 #s: 2, 3, 6

Homework 2, cont.

10. Prove $0 < x < y$ implies $y^{-1} < x^{-1}$.

Suppose $0 < x < y$ is true. Since $0 < x$ and $0 < y$, we can multiply both sides by $x^{-1}y^{-1}$ and preserve the equality. So, $0 \cdot (x^{-1}y^{-1}) < x(x^{-1}y^{-1}) < y(x^{-1}y^{-1})$ and $x^{-1}y^{-1}$ are positive therefore their product is positive.

By 1.3.2, $0(x^{-1}y^{-1}) = 0$

By M2 and M4, $(x^{-1} \cdot x)y^{-1} = (1)y^{-1}$

By M1, M2 and M4, $(y^{-1} \cdot y)x^{-1} = (1)x^{-1}$

So, $0 < (1)y^{-1} < (1)x^{-1}$ which is $0 < y^{-1} < x^{-1}$ by M3. This implies $y^{-1} < x^{-1}$ which is what we wanted to prove. $\square$

4. 1. For each of the following sets, describe the set of all upper bounds for the set:

a) of odd integers;

There are no upper bounds by Def 1.4.7.

b) $\{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{1 - \frac{1}{1} = 0, 1 - \frac{1}{2} = \frac{1}{2}, 1 - \frac{1}{3} = \frac{2}{3}, 1 - \frac{1}{4} = \frac{3}{4}, \dots\}$

So the set is an image under $\mathbb{N}$ of f where $f: \mathbb{N} \rightarrow \mathbb{Q}$ s.t. $f(n) = \frac{n-1}{n}$.

The set of all upper bounds is $[1, \infty)$.

c) $\{r \in \mathbb{Q} : r^3 < 8\}$

The set of all upper bounds is $[\sqrt[3]{8}, \infty)$.

d) $\{\sin x : x \in \mathbb{R}\}$

The set of all upper bounds is $[1, \infty)$.

2. Find the least upper bound of the above sets.

a) None

b) 1

c) 2

d) 1

S.t. $nx < my$.

★ Lemma for Problem 10: $x^{-1}y^{-1} > 0$ iff $x > 0, y > 0$.
or $x < 0, y < 0$. Since it is noted in the problem that
 $0 < x < y$ - I shall only focus on the case where $0 < x, 0 < y$.

Case $x, y > 0$.

$x \cdot x^{-1} \odot 0 \cdot x^{-1}$, By M4, RHS = 1 and by a proven
result, $0 \cdot x = 0$, meaning $1 \odot 0$. We know $>$ is the
only relation that holds, which means $x^{-1} > 0$ or else
the inequality would be reversed.

Homework 2, cont.

7. Prove that if $x < y$ are two real numbers, then there is a rational number r with $x < r < y$. (Use Ex. 1.4.9.)

Proof: For any $x, y \in \mathbb{R}$ where $x < y$,

Then $y - x > 0$. By Ex 1.4.9, $\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < y - x$.

In other words, $1 < ny - nx$. So, the distance between ny and nx is larger than 1 which means there exists $m \in \mathbb{Z}$ s.t. $nx < m < ny$. Hence, $x < \frac{m}{n} < y$, $r = \frac{m}{n} \in \mathbb{Q}$. $\square$

★ Lemma for Problem 10: $x^{-1}y^{-1} > 0$ iff $x > 0, y > 0$ or $x < 0, y < 0$. Since it is noted in the problem that $0 < x < y$ - I shall only focus on the case where $0 < x, 0 < y$.

$x \cdot x^{-1} = 1$, By M4, $RHS = 1$ and by a proven result, $0 \cdot x = 0$, meaning $1 > 0$. We know $>$ is the only relation that holds, which means $x^{-1} > 0$ or else the inequality would be reversed.

1.4 Hs. 10, 12

1.5 Hs. 2, 3, 6

Homework 2, cont.

4 10: Show that if L_x and L_y are Dedekind cuts of $x, y \in \mathbb{R}$, then $L_x + L_y = \{r+s : r \in L_x \text{ and } s \in L_y\}$ is a Dcut.

Proof: $L_x = \{r \in \mathbb{Q} : r < x\}$; Dcut for $x \in \mathbb{R}$
 $L_y = \{s \in \mathbb{Q} : s < y\}$; Dcut for $y \in \mathbb{R}$

Then, $L_x + L_y = \{r+s : r+s < x+y \text{ and } r, s \in \mathbb{Q}\}$
 $L_x + L_y = \{r+s : r \in \mathbb{Q} < x \text{ and } s \in \mathbb{Q} < y\}$
 $L_x + L_y = \{r+s : r \in L_x \text{ and } s \in L_y\}$

to finish this proof, it suffices to show $L_x + L_y$ is itself a Dedekind cut. Therefore, it must satisfy the definition of a Dedekind cut.

① $L_x + L_y \neq \emptyset$ and $L_x + L_y \neq \mathbb{Q}$.

So, $L_x + L_y$ is not $\emptyset$ because, as asserted, L_x and L_y are Dedekind cuts and are therefore not empty. By definition $L_x + L_y = \{r+s : r \in L_x \text{ and } s \in L_y\}$. Since $r \neq s$ and by virtue of L_x, L_y being cuts, then their sum also exists.

Also, were $L_x + L_y = \mathbb{Q}$ then $r+s$ would constitute a minimum element of $\mathbb{Q}$, at maximal values r, s . However,

by Archimedean Property for any $x \in \mathbb{R}$ there is a $n \in \mathbb{N}$ s.t. $x < n$. So, for any $x \in \mathbb{R}$, there is the natural successor of x in $\mathbb{N}$, $S_n(x)$. Since $x, y \in \mathbb{R}$, $S_n(x+y) > x+y$, $S_n(x+y) \in \mathbb{N} > x+y$. So $r+s < x+y < S_n(x+y) \in \mathbb{Q}$ means $L_x + L_y \neq \mathbb{Q}$.

② $L_x + L_y$ has no largest element.

Say it does and call it m . Then $m = r_0 + s_0$ for some maximal elements $r_0 \in L_x$ and $s_0 \in L_y$. But in $L_x + L_y$ doesn't exist because L_x and L_y are Dcuts and therefore have no maximal elements. Hence, by contradiction, $L_x + L_y$ has no largest element.

③ If $a \in (L_x + L_y)$, then so is every $b \in \mathbb{Q}$ with $b < a$.
Say $a \in (L_x + L_y)$. Then, $a \in \mathbb{Q}$ of the form $r+s$ where $r < x$ and $s < y$. So, $a = r+s < x+y$. Now say $b \in \mathbb{Q} < a$. Then $b < a = r+s < x+y$. Since $b \in \mathbb{Q}$ and $b < x+y$, then $b \in L_x + L_y$ of the form $r+s$. So, $L_x + L_y$ satisfies the definition of a Dedekind cut. $\square$

1.4 #5: 12

1.5 #5: 2, 3, 4

Homework 2, cont.

12: If L is the Dedekind cut of example 1.4.2 and L determines the real number x (so that $L = L_x$), prove $L_x = L_2$. Thus the real number corresponding to L has square 2.

$$L = \{r \in \mathbb{Q} : r \geq 0 \text{ and } r^2 < 2\} \cup \{r \in \mathbb{Q} : r < 0\}$$

$$L_{x_1} = \{r \in \mathbb{Q} : r \geq 0 \text{ and } r^2 < 2\} \cup \{r \in \mathbb{Q} : r < 0\}$$

$$L_{x_2} = \{s \in \mathbb{Q} : s \geq 0 \text{ and } s^2 < 2\} \cup \{s \in \mathbb{Q} : s < 0\}$$

$$L_{x_1} \cdot L_{x_2} = \{rs \in \mathbb{Q} : r \in L_{x_1}, r \geq 0, s \in L_{x_2}, s \geq 0\} \cup \{x \in \mathbb{Q} : x < 0\}$$

$$L_2 = \{q \in \mathbb{Q} : q < 2\}$$

Proof: $L_{x_1}, L_{x_2} \subset L_2$

It suffices to show that for every $rs \in L_{x_1}, L_{x_2}$, $rs \in L_2$.

So, $r^2 < 2$ by definition of L_{x_1} and $s^2 < 2$ by definition of L_{x_2} .

Then, $r^2 s^2 < 2 \cdot 2 = 4$; $(rs)^2 < 4$; $2 < rs < 2$.

So, $rs < 2$ implies $rs \in L_2$ since L_{x_1} and L_{x_2} contain elements less than 0 by their unions.

Now to show $L_2 \subset L_{x_1}, L_{x_2}$.

It suffices to show for any $q \in L_2$ that $q \in L_{x_1}, L_{x_2}$.

So, $q < 2$ by definition of L_2 . Then, $-4 < q^2 < 4$.

This is the same as $q^2 < 4$ (while $q \geq 0$).

Then $q_1 \cdot q_2 < 2 \cdot 2$ (while $q \geq 0$ and q_1, q_2 are $q \in L_2$).

Then it is reasonable to say $q_1 < 2$ and $q_2 < 2$.

So $q_1 \in L_{x_1}$ and $q_2 \in L_{x_2}$ and $q_1, q_2 \in L_{x_1}, L_{x_2}$.



1.5 HS: 2, 3, 6

Homework 2, cont.

2. Find the sup and inf of the following sets. Tell whether each set has a max or min.

a) $(-2, 8] = A$. $\sup A = 8$, $\inf A = -2$, $\max = 8$, $\min = N/A$

b) $\left\{ \frac{n+2}{n^2+1} \right\} = A$; $\inf A = 0$ by Archimedean Property, $\sup A = 2$.

$\min = N/A$, $\max = 2$

c) $\{n/m : n, m \in \mathbb{Z}, n^2 < 5m^2\} = A$; $\sup A = \sqrt{5}$, $\inf A = -\sqrt{5}$

$(-\sqrt{5}, \sqrt{5})$; $\max$ & $\min$ do not exist.

3. Prove that if $\sup A < \infty$, then for each $n \in \mathbb{N}$ there is an element $a_n \in A$ s.t. $\sup A - 1/n < a_n \leq \sup A$.

Proof: It is reasonable to rewrite the inequality by adding $\frac{1}{n}$.

So, $\sup A < a_n + \frac{1}{n} \leq \sup A + \frac{1}{n}$.

By definition, $a_n \leq \sup A$ as $\sup A$ may be an element of A so $a_n \in A$ may be the greatest element of A and also, arbitrarily, $a_n < \sup A$ by def 1.5.2.

It naturally follows that $a_n + \frac{1}{n} \leq \sup A + \frac{1}{n}$ for any $a_n \in A$ and $n \in \mathbb{N}$.

It therefore suffices to merely prove: $\sup A < a_n + \frac{1}{n}$ for any n , the existence of a_n .

S: $\forall n \in \mathbb{N} \exists a_n \in A (\sup A - \frac{1}{n} < a_n \wedge a_n \leq \sup A)$

$\sim S$: $\exists n \in \mathbb{N} \forall a_n \in A (\sup A - \frac{1}{n} \geq a_n \vee a_n > \sup A)$

$a_n + \frac{1}{n} \leq \sup A$ which is a contradiction by definition of $\sup A$.

$a_n > \sup A$ is a false statement by def. $\sup A$.

Thus, S must be true.

□

6: Prove: d) $\sup(A-B) = \sup A - \inf B$.

Proof: By definition, $A-B = \{a-b : a \in A, b \in B\}$
 $a \in A$ where $a \leq \sup A$ and $b \in B$ where $b \leq \sup B$.

To rewrite, $\sup(A-B) = \sup(A) + \sup(-B)$ by (c).
Then, $\sup(A) - \inf(B)$ by (b). $\square$